SUMMARY · MK 6 BUSINESS ANALYTICS · WEEK 10

# Advanced Analytics: Big Data Analytics

This week enters the first advanced-analytics topic: big data. Data becomes big data not at a fixed size but when it exceeds what conventional tools (spreadsheets, manual reading) can handle. Its characteristics are the four Vs: volume, velocity, variety and veracity. The internet is the main source, offering speed, low-cost coverage, unprompted customer voice and early warning (examples: Google Flu Trends, the Billion Prices Project, Yelp reviews in Boston, PetaBencana.id). Because internet data is large, unstructured and dirty, it needs different methods. The case is 11,488 posts about BrandX (fictional) in September 2026, analysed with three methods: sentiment analysis (lexicon vs IndoBERT), topic modelling with BERTopic, and social network analysis (in-degree, betweenness, communities). Each result is read against a baseline, over time, with its reasons and its possible errors.

## Material

### 1 What big data is, and the four Vs

**Big data** is data too large, too fast, too varied or too uncertain for conventional tools such as spreadsheets and manual reading.

**Data**

500 survey answers: fit in one spreadsheet, read in under two hours.

vs

**Big data**

2,000,000 posts about one brand in a month: **231 days** of non-stop reading (2,000,000 × 10 seconds = 20 million seconds).

**The key**

Data becomes big data when it exceeds what conventional tools or people can handle, not at a fixed size.

| V | Question | Social-media example | Explanation |
|---|---|---|---|
| **Volume** | How much data is generated? | Millions of posts | One Excel worksheet holds at most 1,048,576 rows. A shopping app with 1 million daily users × 50 clicks = 50 million rows a day ≈ 48 worksheets. |
| **Velocity** | How fast does it arrive? | New posts every second | Complaints rise on day 3; a monthly report sees it 27 days later, social-media data sees it within minutes. |
| **Variety** | How many forms does it take? | Text, images, video | Most big data is unstructured. Reading 10,000 posts × 10 seconds = about 28 hours. |
| **Veracity** | How far can it be trusted? | Bots, spam, duplicates | Internet data must be filtered and checked before analysis. |

| | Structured | Unstructured |
|---|---|---|
| Form | Rows and columns | Free text, images, video, interactions |
| Example | Online orders, page views | Posts, reviews, photos, videos |
| Analysis | Queries and statistics | Text mining, network analysis |

**Four veracity problems** (posts to remove before analysing opinions about BrandX):

**Spam or bot**

"PROMO!!! Click this link, BrandX voucher 1 million" — posted automatically, not by a customer.

**Duplicate**

"BrandX delivery is slow, don't shop here" — the same text posted by 40 accounts.

**Irrelevant**

"Saw the BrandX Band concert last night, great" — same name, different subject.

**Fake review**

"Best product!!! Recommended seller, 5 stars" — written by an account created today.

## 2 The internet as a big-data source: benefits and examples

Every search, click, review and post on the internet leaves a record. Sources: social media (posts, comments, shares), e-commerce (products, prices, orders), online reviews (ratings), online news (articles, comments), search engines (queries) and video platforms (videos, comments). Most are public and can be collected by keyword.

| Benefit | Example | Data → analysis → result | Lesson |
|---|---|---|---|
| **Speed**: signals before official statistics | Google Flu Trends | Millions of US flu-symptom searches → compared with official regional flu reports → flu activity estimated 1–2 weeks early. But in 2013 the estimate was more than double the official figure. | Volume and velocity made it fast; weak veracity made it inaccurate. |
| **Low-cost coverage**: millions of observations without a survey team | Billion Prices Project | Prices of millions of products collected daily from online retailers → daily inflation indexes → in Argentina, online prices showed inflation well above the official rate. | Websites replaced price-survey teams; the index updates daily. |
| **Unprompted voice**: opinions given without being asked | Boston and Yelp | Yelp restaurant reviews in Boston (2015) → models learned which words signal hygiene problems → inspectors visited the likeliest failures first. | Unstructured text analysed at scale guided where inspectors went. |

| | | | |
|---|---|---|---|
| **Early warning**: events reported as they happen | PetaBencana.id | Jakarta residents' social-media flood reports → checked and mapped as they arrive → a real-time flood map for residents and Jakarta's disaster agency. | Reports arrive within minutes, faster than any survey. |

**Why big data needs different methods:**

| | Conventional data | Internet data | Required method |
|---|---|---|---|
| Amount | Hundreds of rows | Millions of posts | Automated processing |
| Structure | Rows and columns | Text and interactions | Text mining, network analysis |
| Quality | Checked when entered | Bots, spam, duplicates | Filtering and checking |

Each post carries two kinds of data: **text** (what is said → text mining) and **interactions** (who responds to whom → network analysis).

Collect: posts matching a keyword

→ Convert: text into numbers a model can read

→ Analyse: sentiment and topics

**Text mining** converts unstructured text into structured data that can be counted and modelled.

## 3 The BrandX case and sentiment analysis

**The case:** BrandX sells consumer products online. Posts mentioning BrandX on X, Instagram, TikTok and Google Maps were collected by keyword for September 2026: **11,488 posts** in Indonesian. The brand, accounts and all data are fictional.

| Method | Question |
|---|---|
| Sentiment analysis | How do customers feel? |
| Topic modelling (BERTopic) | What do they talk about? |
| Social network analysis | Who spreads the conversation? |

**Motivation:** reading 11,488 posts × 10 seconds = **32 hours**. Sentiment analysis labels every post automatically: **positive, neutral or negative**.

| Post | Sentiment |
|---|---|
| Kurirnya ramah, paket sampai dengan aman (friendly courier, parcel arrived safely) | Positive |
| Estimasi pengiriman ke Medan berapa hari ya? (how many days to Medan?) | Neutral |
| Chat CS dari kemarin gak dibalas (CS hasn't replied since yesterday) | Negative |

### Lexicon-based

Sums the scores of words in a sentiment dictionary. **Strength:** transparent, no labelled data needed. **Weakness:** misses negation, sarcasm and slang.

vs

### Machine learning

A model learns from posts labelled by people. **Strength:** reads each word in the context of the sentence. **Weakness:** needs labelled data; harder to explain.

**Lexicon scoring:** bagus (good) +1, cepat (fast) +1, ramah (friendly) +1, lambat (slow) −1, rusak (broken) −1, kecewa (disappointed) −1. Score > 0 = positive, 0 = neutral, < 0 = negative.

| Post | Score | Label |
|---|---|---|
| pengiriman cepat dan barang bagus | +1 +1 = +2 | Positive |
| pengiriman cepat tapi barangnya rusak | +1 −1 = 0 | Neutral (a human would say negative) |
| promonya bagus, pengirimannya cepat, tapi cs lambat | +1 +1 −1 = +1 | Positive |

| Lexicon limit | Example | Wrong result |
|---|---|---|
| Negation | pelayanannya tidak ramah (service not friendly) | +1 → positive |
| Sarcasm | mantap, paketnya cuma telat seminggu (great, only a week late) | 0 → neutral |
| Slang | pengirimannya cpt bgt (delivery super fast) | 0 → "cpt" isn't in the dictionary |

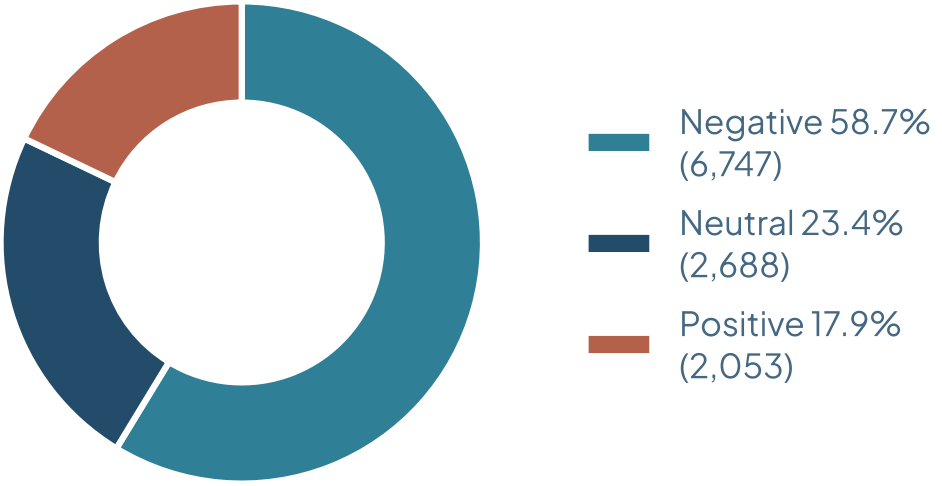
A lexicon reads words one at a time, not context. So **IndoBERT** is used:

1. **Model:** a transformer language model pre-trained on a large collection of Indonesian text.
2. **Fine-tuning:** trained further on posts labelled positive, neutral or negative.
3. **Prediction:** for a new post it returns a label with a probability, reading each word in the context of the sentence.

| Post | IndoBERT label | Probability |
|---|---|---|
| CS @BrandX lambat banget balasnya | Negative | 0.95 |
| Barang sampai aman, makasih @BrandX | Positive | 0.97 |
| @BrandX ada promo akhir bulan gak? | Neutral | 0.92 |
| mantap, paketnya cuma telat seminggu | Negative | 0.64 |

The sarcastic post the lexicon scored 0 is now labelled negative, though with a lower probability (0.64).

**Sentiment of 11,488 BrandX posts (IndoBERT)**




More than half the posts are negative. Whether that's high or low only means something against a baseline.

**Result:** negative **58.7%** (6,747), neutral 23.4% (2,688), positive 17.9% (2,053). More than half the posts are negative.

| How to read it | Explanation |
|---|---|
| Baseline | 58.7% negative is high or low only against something: other days, other months, or a competitor. |
| Trend | A rising negative share shows when a complaint grows. On 18 September: 75% negative vs 54% on other days → when a complaint became a crisis. |
| Reasons | Sentiment shows feelings, not causes. Causes come from topic modelling. |
| Errors | Labels are predictions. Check a sample of posts, starting with low-probability ones. |

! **What sentiment alone can't show**

That most complaints are about delivery (needs topics), that sales fell (no sales data), or that all negative posts came from real customers (needs

veracity checks). What it can show: 18 September was more negative than usual.

## 4 Topic modelling with BERTopic

**Motivation:** there are 6,747 negative posts. Which problems do customers raise, which most often, and which to fix first? Topic modelling finds the themes without reading every post.

**Topic modelling** is an *unsupervised* method that groups documents into topics by content similarity. No labels needed: the topics are found in the data. Example: "parcel not arrived, courier slow" and "tracking number hasn't moved since Monday" → topic 1 (parcel, courier, tracking, arrived); "refund not in my balance yet" and "return money promised in three days" → topic 2 (refund, money, return, balance).

Embed: posts become embeddings → Reduce: UMAP reduces dimensions → Cluster: HDBSCAN groups similar posts → Describe: c-TF-IDF finds typical words

Posts that fit no cluster are classified as **outliers**.

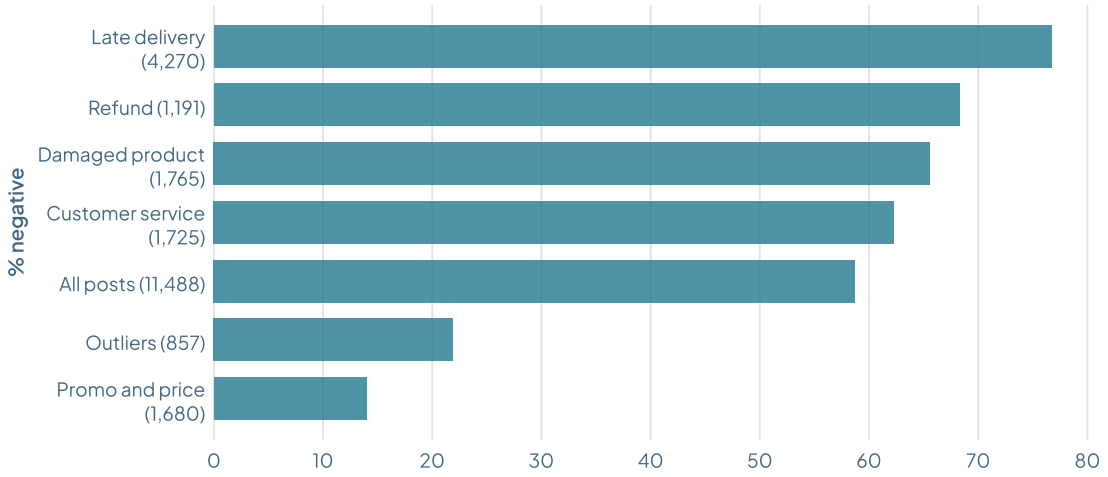
**What is an embedding?**
A vector of numbers representing a post's *meaning*. Posts with similar meaning sit close together, even with different words. Example: "item broken when opened" and "product defective, not like the photo" are close (both damaged product), while "admin hasn't replied since morning" is far away (customer service).

| Topic | Typical words | Posts | Share | Negative |
|---|---|---|---|---|
| Late delivery | kirim, telat, kurir, paket, resi | 4,270 | 37.2% | 76.7% |
| Damaged product | rusak, barang, sesuai, cacat, foto | 1,765 | 15.4% | 65.6% |

| | | | | |
|---|---|---|---|---|
| Customer service | cs, balas, chat, lambat, admin | 1,725 | 15.0% | 62.3% |
| Promo and price | promo, diskon, voucher, harga, murah | 1,680 | 14.6% | 14.0% |
| Refund | refund, dana, retur, proses, saldo | 1,191 | 10.4% | 68.3% |
| Outliers | no single theme | 857 | 7.5% | 21.9% |

Topic names are assigned by the analyst after reading the typical words. Each post belongs to one topic or to the outliers.

**Negative share by topic (%)**




Four topics sit above the 58.7% average. Late delivery is the largest and most negative.

| How to read it | Explanation |
|---|---|
| Size | The share of posts shows how widely a problem is discussed. Late delivery is 37.2%. |
| Sentiment | Compare each topic's negative share with 58.7% (all posts) to see which problem hurts the brand most. |
| Names | A topic name is the analyst's reading; check it against example posts. |
| Outliers | The 7.5% outside every topic may hold a new issue that is still small. |

## Conclusion

Four of five topics are more negative than the conversation overall. Late delivery is the largest *and* most negative topic: the first problem to address. Promo and price is far more positive (14% negative).

## 5 Social network analysis

**Motivation:** sentiment and topics show *what* is said, not *who* spreads it. Example: u07's post "Parcel not arrived, @BrandX what's up?" was retweeted or mentioned by 0 accounts; the Influencer's similar post by 10. Who spreads complaints furthest? Who connects separate groups? Do the brand's replies reach its critics?

| Element | Meaning |
|---|---|
| **Node** | An account: a customer, the brand, an influencer or a news outlet |
| **Edge** | A retweet, mention or reply, directed from the account that responds to the account that posted |

| Measure (from Week 9) | Meaning |
|---|---|
| **In-degree** | How often an account is retweeted or mentioned |
| **Betweenness** | How often an account lies on the shortest paths between other accounts (acts as a bridge) |
| **Community** | A group of accounts that interact mostly with each other |

**The BrandX retweet network:** 23 accounts, 29 directed edges (node size = in-degree; users anonymised).

| Account | In-degree | Betweenness |
|---|---|---|
| Influencer | **10** | 0.57 |
| BrandX | 7 | 0.44 |
| News | 6 | **0.68** |

**Communities** (Leiden algorithm, modularity 0.50): community 1 = 9 of 10 accounts negative (critics, around the Influencer); community 2 = 7 of 8 positive (supporters, around BrandX); community 3 = 5 of 5 neutral (news readers).

| How to read it | Explanation |
|---|---|
| In-degree | Influencer (10) is retweeted most: its complaints reach the most accounts. |
| Betweenness | News (0.68) connects the groups: information passes between communities through it. |
| Communities | Critics, supporters and news readers form separate groups that see different posts. |
| Missing edges | No direct edge links BrandX to the critical community: BrandX's replies don't reach them. |

! **Business implication**

BrandX only "talks" to its own supporters. The only path to the critics runs through News. Sensible steps: reply directly to the Influencer's and key critics' complaints, and brief the news outlets as a bridge.

**Practice in SocialX Studio** (socialx.id): (1) choose a data source (X, Instagram, TikTok or online news); (2) enter a keyword, e.g. a brand name; (3) collect matching posts; (4) run sentiment analysis (IndoBERT); (5) run topic modelling (BERTopic); (6) run social network analysis.

## Key takeaways

1. Big data = data that exceeds what conventional tools or people can handle, not a fixed size.
2. Four Vs: volume (amount), velocity (speed), variety (forms), veracity (trustworthiness).
3. One Excel worksheet = 1,048,576 rows; 50 million clicks a day ≈ 48 worksheets.
4. Most big data is unstructured (text, images, video, interactions) → needs text mining and network analysis.
5. Veracity problems: spam/bots, duplicates, irrelevant posts, fake reviews → filter before analysing.
6. Benefits of internet data: speed (Google Flu Trends), low-cost coverage (Billion Prices Project), unprompted voice (Yelp Boston), early warning (PetaBencana.id).
7. Google Flu Trends: fast thanks to volume and velocity, inaccurate due to weak veracity (2013 over double the official figure).
8. Sentiment: lexicon (transparent, fails on negation, sarcasm, slang) vs machine learning such as IndoBERT (reads context, needs labelled data).
9. BrandX: 58.7% negative; read it with a baseline, trend (18 September 75%), reasons (topics) and errors (check samples).
10. BERTopic: embed → UMAP → HDBSCAN → c-TF-IDF; unsupervised; outliers may hold new issues.
11. Late delivery: the largest (37.2%) and most negative (76.7%) topic → the first priority.
12. SNA: node = account, edge = retweet/mention/reply. In-degree = reach, betweenness = bridge, community = group. BrandX isn't directly linked to its critics.

## Case studies

Case & worked answers

## Practice: from 11,488 posts to a BrandX action plan

*A practice combining the three methods; figures from the slides, recommendations are analysis.* BrandX's board asks: "What should we do this week?"

1. **Clean the data (veracity):** remove spam/bots, duplicates (e.g. the same text from 40 accounts), irrelevant posts (BrandX Band, BrandX FC) and fake reviews before counting anything.
2. **Sentiment:** 58.7% negative. Compare with last month's baseline or a competitor. The 18 September spike (75% vs 54%) marks when a complaint became a crisis; read that day's posts to find the trigger.
3. **Topics:** late delivery is the largest (37.2%) and most negative (76.7%) → priority 1. Refund (68.3%) and damaged product (65.6%) next. Promo and price is fine (14%). Check the 7.5% outliers for new issues.
4. **Network:** the Influencer (in-degree 10) spreads complaints furthest; News (betweenness 0.68) is the bridge; there's no BrandX edge to the critical community.
5. **Action plan:** (a) fix courier SLAs and send proactive tracking updates; (b) speed up refunds and publish a target time; (c) have CS reply directly to the Influencer's and critics' posts, not just supporters; (d) brief the news outlets.
6. **Re-measure:** track the delivery topic's negative share and whether BrandX has a direct edge to the critical community next month.

## Worked examples

Problem 1 · Easy **Volume: how many worksheets?**

A shopping app has 1 million daily users, each producing 50 clicks. One Excel worksheet holds at most 1,048,576 rows. How many worksheets for one day, and for one month (30 days)?

1. ```
   Worksheets = rows ÷ 1,048,576
   ```
2. Rows per day = 1,000,000 × 50 = 50,000,000
3. Worksheets per day = 50,000,000 ÷ 1,048,576 = 47.7 → 48
4. Per month = 50,000,000 × 30 = 1.5 billion rows ÷ 1,048,576 = 1,431 worksheets

**Answer:** About 48 worksheets a day; about 1,431 a month. Clearly needs big-data tools, not a spreadsheet.
   ```

Problem 2 Easy **Manual reading time**

How long to read (a) 10,000 posts, (b) BrandX's 11,488 posts and (c) 2,000,000 posts, at 10 seconds per post with no breaks?

1. `Time = posts × 10 seconds`
2. (a) 10,000 × 10 = 100,000 s ÷ 3,600 = 27.8 h ≈ 28 hours
3. (b) 11,488 × 10 = 114,880 s ÷ 3,600 = 31.9 h ≈ 32 hours
4. (c) 2,000,000 × 10 = 20,000,000 s ÷ 86,400 = 231 days

**Answer:** About 28 hours, 32 hours and 231 days. That's why automatic labelling is needed.

Problem 3 Easy **Lexicon sentiment score**

Lexicon: bagus +1, cepat +1, ramah +1, lambat −1, rusak −1, kecewa −1. Score and label: (a) "kurirnya ramah tapi barangnya rusak, kecewa"; (b) "pengiriman cepat, barang bagus, cs ramah"; (c) "pelayanannya tidak ramah". Which does the lexicon misread?

1. ```
   Score = Σ word scores · > 0 positive, 0 neutral, < 0 negative
   ```
2. (a) ramah +1, rusak −1, kecewa −1 = −1 → negative
3. (b) cepat +1, bagus +1, ramah +1 = +3 → positive
4. (c) ramah +1 = +1 → positive, though it means negative (the negation "tidak" isn't read)

**Answer:** (a) −1 negative; (b) +3 positive; (c) +1 positive but wrong: an example of the negation weakness.
   ```

Problem 4 Medium **Negative share and count by topic**

From BrandX's topic table, compute the number of negative posts for late delivery (4,270; 76.7%), refund (1,191; 68.3%) and promo (1,680; 14.0%). Then check: does the total across all topics match 6,747?

1. `Negative posts = topic posts × negative share`
2. Delivery: 4,270 × 0.767 = 3,275
3. Refund: 1,191 × 0.683 = 813; Promo: 1,680 × 0.140 = 235
4. Others: damaged 1,765 × 0.656 = 1,158; CS 1,725 × 0.623 = 1,075; outliers 857 × 0.219 = 188
5. Total = 3,275 + 1,158 + 1,075 + 235 + 813 + 188 = 6,744 ≈ 6,747 (difference from rounding)

**Answer:** Delivery 3,275 negative (almost half of all negative posts); the total matches about 6,747.

Problem 5 Medium **Computing in-degree**

Directed retweet edges (from retweeter to poster): A→C, B→C, D→C, E→C, C→F, D→F, E→D. Compute each account's in-degree. Which account has the most reach?

1. ```
   In-degree = number of arrows pointing into the account
   ```
2. C: from A, B, D, E = 4
3. F: from C, D = 2
4. D: from E = 1
5. A, B, E: 0

**Answer:** C has the highest in-degree (4): retweeted most, widest reach. C also links A/B to F, so it may act as a bridge.

## Practice questions

Try answering first. The answer key is below each question.

**1. Data becomes big data when…**

A. It exceeds 1 million rows
B. It exceeds what conventional tools or people can handle
C. It's stored in the cloud
D. It comes from the internet

**Answer: B.** Not a fixed size, but beyond conventional tools.
   ```

**2. Which no longer fits in one Excel worksheet (1,048,576 rows)?**

A. 300 employee records
B. 50,000 product reviews
C. 2,000,000 posts about a brand
D. 1,000 survey responses

**Answer: C.** 2 million > 1,048,576 rows.

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**3. Which must be analysed within minutes (velocity)?**

A. Last year's annual report
B. A complaint spreading on social media
C. Quarterly sales figures
D. A yearly customer survey

**Answer: B.** A complaint can become a crisis within hours.

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**4. Which is unstructured data?**

A. Online orders: number, product, price
B. Product reviews on a marketplace
C. Daily page views per page
D. A warehouse stock table

**Answer: B.** Reviews and buyer photos have no predefined rows and columns.

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**5. "Free IDR 100k credit! Follow and RT this account #BrandX" should be removed because of…**

A. Volume
B. Velocity
C. Variety
D. Veracity (spam/bot)

**Answer: D.** It's not a customer opinion about BrandX.

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**6. The lesson of Google Flu Trends is…**

A. Big data is always accurate
B. Fast thanks to volume and velocity, but inaccurate due to weak veracity
C. Searches are unrelated to flu
D. Official data is always a year late

**Answer: B.** In 2013 the estimate was more than double the official figure.

---

**7. PetaBencana.id is an example of which benefit of internet data?**

A. Low-cost coverage
B. Early warning
C. Unprompted voice
D. Tax savings

**Answer: B.** Residents' flood reports arrive within minutes and are mapped in real time.

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**8. The Billion Prices Project showed that…**

A. Online prices are always cheaper
B. Websites can replace price surveys, and Argentina's online inflation was well above the official rate
C. Inflation can't be measured
D. Manual surveys are faster

**Answer: B.** An example of low-cost coverage.

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**9. A weakness of lexicon-based sentiment analysis is…**

A. Needs labelled data
B. Misses negation, sarcasm and slang
C. Hard to explain
D. Too slow

**Answer: B.** A lexicon reads words one at a time, not context.

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**10. Lexicon: bagus +1, cepat +1, lambat −1. "promonya bagus, pengirimannya cepat, tapi cs lambat" scores…**

A. −1, negative
B. 0, neutral
C. +1, positive
D. +3, positive

**Answer: C.** +1 +1 −1 = +1 > 0 → positive.

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**11. "pelayanannya tidak ramah" (service not friendly) is labelled positive by the lexicon because of…**

A. Sarcasm
B. Unread negation
C. Slang
D. Duplication

**Answer: B.** "ramah" counts +1; "tidak" (not) is ignored.

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**12. "Great @BrandX, CS replied after three days." The right label…**

A. Positive
B. Neutral
C. Negative (sarcasm)
D. Can't be labelled

**Answer: C.** "Great" is sarcastic; a contextual model like IndoBERT is more likely to catch it.

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**13. The IndoBERT step that trains the model further on labelled posts is…**

A. Pre-training
B. Fine-tuning
C. Clustering
D. Embedding

**Answer: B.** Pre-training on general Indonesian text; fine-tuning on sentiment-labelled posts.

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**14. Which conclusion does the sentiment result alone support?**

A. Most complaints were about delivery
B. 18 September was more negative than usual
C. Sales fell on 18 September
D. All negative posts came from real customers

**Answer: B.** Causes need topics; sales need other data; authenticity needs veracity checks.

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**15. The BERTopic steps are…**

A. Cluster → embed → describe → reduce
B. Embed → reduce (UMAP) → cluster (HDBSCAN) → describe (c-TF-IDF)
C. Label → train → predict
D. Collect → label → score

**Answer: B.** Posts that fit no cluster become outliers.

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**16. Topic modelling is a…**

A. Supervised method: needs labels
B. Unsupervised method: topics are found in the data
C. Lexicon-based method
D. Network analysis

**Answer: B.** The analyst names topics after reading the typical words.

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**17. The two posts closest in meaning (embedding space)…**

A. "item broken when opened" and "admin not replying"
B. "item broken when opened" and "product defective, not like the photo"
C. "admin not replying" and "product defective"
D. All equally close

**Answer: B.** Same meaning (damaged product) despite different words.

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**18. "Refund promised in 3 working days, now it's 2 weeks" belongs to which topic and sentiment?**

A. Promo and price, positive
B. Refund, negative
C. Late delivery, neutral
D. Outliers, neutral

**Answer: B.** Typical words: refund, money; a broken-promise complaint.

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**19. Which topic should BrandX address first?**

A. Promo and price
B. Late delivery: largest (37.2%) and most negative (76.7%)
C. Outliers
D. Customer service

**Answer: B.** Combine size and negative share.

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**20. In a retweet network, in-degree measures…**

A. How often an account retweets others
B. How often an account is retweeted or mentioned
C. How often an account is a bridge
D. The number of communities

**Answer: B.** The Influencer has the highest in-degree (10).

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**21. News has the highest betweenness (0.68). This means…**

A. It's retweeted most
B. It's the main bridge: information between communities passes through it
C. It's the most negative
D. It's a bot

**Answer: B.** Betweenness = often on the shortest paths between other accounts.

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**22. The "missing edges" finding in the BrandX network is…**

A. BrandX has no followers
B. No direct edge links BrandX to the critical community, so its replies don't reach them
C. News is connected to nobody
D. The Influencer is inactive

**Answer: B.** The only path to the critics runs through News.

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### Sources

- P3MD MK6 Week 10 — Business Analytics: Big Data Analytics (fictional BrandX case; SocialX)
- Slide examples: Google Flu Trends; Billion Prices Project; Yelp and Boston restaurant inspections (2015); PetaBencana.id; IndoBERT; BERTopic